CCNA Practice Exam – Group 1-4 (1–200)

CCNA Practice Exam – Group 1 (1–50)

Network Fundamentals & Access

1. Which layer of the OSI model is responsible for establishing, managing, and terminating sessions between applications?
   A. Session
   B. Application
   C. Transport
   D. Presentation
2. A router interface is configured with IP address 192.168.1.1 /24. Which of the following host IPs would *not* be valid on that subnet?
   A. 192.168.1.2
   B. 192.168.1.100
   C. 192.168.1.255
   D. 192.168.1.10
3. Which two statements describe TCP? *(Choose two)*
   A. Connection-oriented
   B. Uses sequence numbers
   C. Unreliable transport
   D. No acknowledgments
   E. Uses UDP port 80
4. Which command verifies all interfaces' IP addresses and operational status on a Cisco device?
   A. show ip route
   B. show interfaces
   C. show ip interface brief
   D. show version
5. What is the function of ARP in an Ethernet network?
   A. Resolves hostnames to IP addresses
   B. Resolves IP addresses to MAC addresses
   C. Translates private addresses to public
   D. Discovers neighboring routers
6. A switch port configured for **VLAN 20** receives untagged traffic. What will the switch do?
   A. Drop the frame
   B. Forward to native VLAN
   C. Assign it to VLAN 20
   D. Tag it with VLAN 1

7. Which command enables trunking on an interface without negotiation?
   A. switchport mode trunk
   B. switchport mode dynamic desirable
   C. switchport nonegotiate
   D. switchport mode access
8. Which two protocols are used by IPv6 for neighbor discovery and address resolution? *(Choose two)*
   A. ARP
   B. ICMPv6
   C. ND
   D. DHCPv6-PD
   E. EUI-64
9. Which cable type supports 1 Gbps for 100 meters?
   A. Cat 3
   B. Cat 5e
   C. Cat 6 A
   D. Coaxial
10. Which wireless standard operates only in the 5 GHz band?
    A. 802.11g
    B. 802.11b
    C. 802.11a
    D. 802.11n
11. What does a switch use to forward Ethernet frames?
    A. IP address
    B. MAC address
    C. ARP table
    D. Routing table
12. Which command creates VLAN 30 on a Cisco switch?
    A. vlan database 30
    B. create vlan 30
    C. vlan 30
    D. switchport vlan 30
13. What is the function of a default gateway on a host?
    A. Resolves DNS
    B. Provides DHCP leases
    C. Routes packets outside the local network
    D. Stores MAC-address mappings
14. Which IP address block is reserved for Automatic Private IP Addressing (APIPA)?
    A. 169.254.0.0/16
    B. 10.0.0.0/8
    C. 172.16.0.0/12
    D. 192.168.0.0/16

15. You connect two switches back-to-back using a trunk. VLAN 10 and 20 should pass, VLAN 1 should not. Which command completes this?

```
switchport trunk allowed vlan 10,20
```

A. Correct configuration
B. Must include “1,10,20”
C. Wrong syntax
D. Needs “switchport mode access”

16. A PC shows “Media disconnected.” Which OSI layer do you troubleshoot first?
    A. Physical
    B. Data link
    C. Network
    D. Transport
17. What are the non-overlapping 2.4 GHz channels used in most WLANs?
    A. 1, 3, 6
    B. 1, 6, 11
    C. 2, 5, 9
    D. 1, 4, 8
18. Which two commands verify trunk status and allowed VLANs? *(Choose two)*
    A. show interfaces trunk
    B. show vlan brief
    C. show mac address-table
    D. show interfaces status
19. Which STP feature prevents another switch from becoming root on an access port?
    A. BPDU Guard
    B. Root Guard
    C. PortFast
    D. Loop Guard
20. A switch interface is err-disabled after connecting a new PC. Which feature likely caused this?
    A. Port security violation
    B. Spanning-tree block
    C. Duplex mismatch
    D. DHCP snooping
21. What is the administrative distance of a directly connected route?
    A. 0
    B. 1
    C. 90
    D. 110
22. Which two OSPF router types can generate LSAs? *(Choose two)*
    A. ABR

B. ASBR
C. DR
D. BDR

23. What is the default OSPF Hello timer on broadcast networks?
    A. 5 s
    B. 10 s
    C. 30 s
    D. 60 s
24. Which routing protocol uses multicast address 224.0.0.5?
    A. RIP
    B. EIGRP
    C. OSPF
    D. HSRP
25. Which command creates a static route to 10.10.10.0/24 via 192.168.1.2?
    A. ip route 10.10.10.0 255.255.255.0 192.168.1.2
    B. route add 10.10.10.0 mask 255.255.255.0 192.168.1.2
    C. ip static 10.10.10.0 255.255.255.0 192.168.1.2
    D. set ip route 10.10.10.0 192.168.1.2
26. Which command shows OSPF neighbor relationships?
    A. show ip ospf neighbor
    B. show ip route ospf
    C. show protocols ospf
    D. show ip ospf interface
27. What is the purpose of the “standby preempt” command in HSRP?
    A. Forces the router to always be standby
    B. Allows higher-priority router to reclaim active role
    C. Prevents state change
    D. Enables tracking
28. Which protocol provides default-gateway redundancy across multiple routers?
    A. HSRP
    B. STP
    C. LACP
    D. VTP
29. What NAT type allows many internal hosts to share one public IP?
    A. Static NAT
    B. Dynamic NAT
    C. PAT
    D. Bidirectional NAT
30. Which command enables NAT inside interface?
    A. ip nat inside
    B. ip nat enable

C. nat inside enable
D. ip nat source

31. What is the purpose of the “ip helper-address” command?
    A. Relay DHCP requests
    B. Enable DNS caching
    C. Redirect ICMP packets
    D. Advertise default gateway
32. Which DHCP message does a client send to request an offered IP?
    A. Discover
    B. Request
    C. Offer
    D. Acknowledge
33. What syslog severity level indicates a critical condition?
    A. 0
    B. 1
    C. 2
    D. 3
34. Which SNMP version supports authentication and encryption?
    A. v1
    B. v2c
    C. v3
    D. v2
35. What is the default logging buffer size in IOS (approx.)?
    A. 4096 bytes
    B. 8192 bytes
    C. 16 KB
    D. 32 KB
36. Which command sends logs to a syslog server at 10.0.0.10?
    A. logging host 10.0.0.10
    B. log server 10.0.0.10
    C. syslog 10.0.0.10
    D. set logging server 10.0.0.10
37. What ACL statement permits HTTP traffic from 10.0.0.0/8 to 192.168.1.10?
    A. access-list 101 permit tcp 10.0.0.0 0.255.255.255 host 192.168.1.10 eq 80
    B. permit ip 10.0.0.0 host 192.168.1.10
    C. access-list 101 permit udp any any eq 80
    D. access-list 101 permit tcp any any eq 443
38. Which feature prevents rogue DHCP servers?
    A. Port security
    B. DHCP snooping

C. ARP inspection
D. Storm control

39. Which command enables port security on interface Gi0/1?
A. switchport port-security
B. enable port security
C. switchport mac-secure
D. mac-guard enable

40. Which command limits maximum 2 MAC addresses on a port?
A. switchport port-security maximum 2
B. mac limit 2
C. port-security limit 2
D. switchport mac count 2

41. What does “sticky” do in port-security configuration?
A. Learns and saves MACs dynamically to running config
B. Locks port to one static MAC only
C. Erases learned addresses on reboot
D. Allows unlimited addresses

42. Which feature validates ARP replies using DHCP snooping binding?
A. ARP inspection
B. Port security
C. BPDU guard
D. Loop guard

43. Which command enables SSH access only on VTY 0–4?
A. line vty 0 4 → transport input ssh
B. enable ssh vt
C. ip ssh on
D. crypto ssh enable

44. What port does SSH use?
A. 21
B. 22
C. 23
D. 443

45. Which algorithm does WPA2 use for encryption?
A. TKIP
B. AES-CCMP
C. RC4
D. DES

46. Which three AAA components are correct? *(Choose three)*
A. Authentication
B. Authorization

C. Accounting
D. Allocation

47. Which IPsec protocol provides data confidentiality?
    A. AH
    B. ESP
    C. IKE
    D. GRE
48. Which statement best describes REST API?
    A. Uses SNMP for automation
    B. Stateless, uses HTTP verbs and JSON/XML
    C. Uses CLI scripting
    D. Requires agents on switches
49. Which HTTP method is used to *retrieve* data from a REST endpoint?
    A. GET
    B. POST
    C. PUT
    D. DELETE
50. In JSON, what notation defines a list or array?
    A. {}
    B. []
    C. ()
    D. <>

CCNA Practice Exam – Group 2 (51 – 100)

Network Access, IP Connectivity, and IP Services

51. Which protocol discovers directly connected Cisco devices and displays platform and IP information?
    A. LLDP
    B. STP
    C. CDP
    D. VTP
52. A switch shows “Fa0/1 is up, line protocol is down.” What layer should be checked next?
    A. Layer 1
    B. Layer 2
    C. Layer 3
    D. Layer 4
53. Which feature allows multiple switches to appear as one logical switch to downstream devices?
    A. EtherChannel
    B. StackWise
    C. PortFast
    D. VTP

54. Which command removes VLAN 20 from a switch?
    A. no vlan 20
    B. delete vlan 20
    C. remove vlan 20
    D. erase vlan 20

55. Which command verifies VLAN membership of interface Gi0/2?
    A. show interfaces trunk
    B. show interfaces switchport
    C. show vlan brief
    D. show mac address-table

56. What occurs if two switches have the same STP priority and MAC address?
    A. One becomes root automatically
    B. Manual configuration required
    C. The lower-numbered port wins
    D. STP fails

57. A switch port shows “blocking” state. What is its function?
    A. Forward frames
    B. Learn MAC addresses
    C. Prevent loops
    D. Drop BPDUs

58. What does PVST+ stand for?
    A. Per VLAN Spanning Tree Plus
    B. Partial VLAN Switch Trunk
    C. Port VLAN Secure Transport
    D. Private VLAN Segmentation Type

59. A switch has a port configured as:

```
interface Gi0/1 switchport mode trunk switchport trunk allowed vlan 10,30
```

Which VLANs will pass?
A. 10 only
B. 30 only
C. 10 and 30
D. All VLANs

60. Which command disables DTP negotiation on a trunk port?
    A. switchport nonegotiate
    B. switchport no-dtp
    C. disable dtp
    D. no switchport mode dynamic

61. What is the default STP priority value?
    A. 0

B. 32768
C. 4096
D. 65535

62. What protocol automatically assigns IP addresses to network devices?
A. DNS
B. NTP
C. DHCP
D. TFTP

63. Which command configures a DHCP relay agent?
A. ip helper-address 10.0.0.5
B. ip dhcp relay 10.0.0.5
C. ip forward 10.0.0.5
D. relay dhcp server 10.0.0.5

64. Which DNS record type maps a hostname to an IPv4 address?
A. PTR
B. A
C. AAAA
D. CNAME

65. Which command shows learned dynamic NAT translations?
A. show ip nat translations
B. show ip route nat
C. show nat session
D. show ip nat details

66. What is the main function of NTP?
A. Name resolution
B. Time synchronization
C. DHCP address leasing
D. Routing updates

67. What port does TFTP use?
A. 20
B. 21
C. 23
D. 69

68. Which statement about SNMP is correct?
A. Uses TCP port 25
B. Used for configuration backup
C. Allows monitoring and alerting of network devices
D. Encrypts all traffic by default

69. Which command configures a router as an NTP client?
A. ntp server 192.0.2.1
B. ip ntp 192.0.2.1

C. set ntp client 192.0.2.1
D. clock update ntp 192.0.2.1

70. A switch port should automatically power an IP phone. Which feature enables this?
    A. LLDP-MED
    B. PoE
    C. AutoQoS
    D. PortFast
71. Which protocol is responsible for segmenting data and ensuring delivery?
    A. IP
    B. UDP
    C. TCP
    D. ICMP
72. Which two protocols operate at the Network layer? *(Choose two)*
    A. IP
    B. ICMP
    C. TCP
    D. ARP
73. Which IP address belongs to a Class B private range?
    A. 10.1.1.1
    B. 172.16.5.5
    C. 192.168.10.10
    D. 169.254.0.5
74. Which command assigns an IPv6 address to an interface?
    A. ipv6 address 2001:db8::1/64
    B. ip address 2001:db8::1 64
    C. ipv6 enable 2001:db8::1/64
    D. ip v6 addr 2001:db8::1/64
75. What is the purpose of link-local IPv6 addresses?
    A. Internet routing
    B. Communication within a single link segment
    C. Static addressing only
    D. Reserved for NAT64
76. Which IPv6 prefix represents unique local addresses (ULA)?
    A. FE80::/10
    B. FC00::/7
    C. FF00::/8
    D. 2001::/16
77. Which command enables IPv6 routing on a Cisco router?
    A. ipv6 unicast-routing
    B. ip routing v6

C. enable ipv6
D. ip unicast-v6

78. Which field in a routing table determines the trustworthiness of a route?
    A. Metric
    B. Next Hop
    C. Administrative Distance
    D. Prefix Length

79. Which command displays OSPF-learned routes?
    A. show ip route ospf
    B. show ip ospf neighbors
    C. show ip ospf interface
    D. show routing ospf

80. What type of OSPF router connects two areas?
    A. Internal
    B. Backbone
    C. ABR
    D. ASBR

81. What is the metric used by OSPF?
    A. Hop count
    B. Bandwidth cost
    C. Delay
    D. Reliability

82. Which of the following represents the default OSPF process ID range?
    A. 1 – 65 535
    B. 0 – 255
    C. 100 – 200
    D. 1 – 4096

83. Which protocol provides neighbor discovery for IPv6?
    A. EIGRPv6
    B. NDP
    C. ARP
    D. BGP

84. Which FHRP is an IETF standard?
    A. HSRP
    B. GLBP
    C. VRRP
    D. IRDP

85. A router with HSRP priority 110 fails and recovers. Preemption is disabled. What happens?
    A. It becomes active again
    B. It remains standby

C. Both become active

D. HSRP stops

86. What does “show standby brief” display?

A. Routing protocol status

B. HSRP state and priority

C. Spanning-tree root

D. EtherChannel summary

87. What port does HSRP use for hello messages?

A. UDP 1985

B. UDP 1986

C. TCP 179

D. UDP 67

88. Which NAT type uses a pool of global addresses for internal users?

A. Static NAT

B. Dynamic NAT

C. PAT

D. Bidirectional NAT

89. Which command shows the number of active NAT translations?

A. show ip nat statistics

B. show ip nat count

C. show ip nat table

D. show ip session nat

90. Which type of NAT mapping is one-to-one?

A. Static

B. Dynamic

C. PAT

D. Overload

91. Which command configures NAT overload using interface Gig0/1?

A. ip nat inside source list 1 interface Gig0/1 overload

B. nat list 1 Gig0/1 overload

C. ip nat overload Gig0/1 list 1

D. ip nat interface Gig0/1 inside overload

92. What is the range of syslog severity levels?

A. 0 – 5

B. 0 – 6

C. 0 – 7

D. 1 – 8

93. What severity level is “Informational”?

A. 5

B. 6

C. 7
D. 4

94. Which command limits log severity to warning and higher?
A. logging trap warnings
B. logging trap warning
C. logging trap 5
D. logging trap 4

95. Which syslog facility code represents local use?
A. local0 – local7
B. system
C. daemon
D. kernel

96. Which ACL type filters traffic based on source and destination ports?
A. Standard
B. Extended
C. Named
D. Dynamic

97. Which command applies ACL 105 outbound on Gig0/0?
A. ip access-group 105 out
B. access-group 105 output
C. ip access-list 105 out
D. access-list 105 out

98. What is the last rule of every ACL if not explicitly defined?
A. Permit all
B. Deny all
C. Deny TCP only
D. Permit ICMP

99. Which security feature prevents spoofed DHCP servers on access ports?
A. Port security
B. DHCP snooping
C. Dynamic ARP Inspection
D. BPDU guard

100. Which DAI command enables validation on VLAN 10?
A. ip arp inspection vlan 10
B. arp inspection enable 10
C. ip arp validate 10
D. ip arp snoop 10

🧭 CCNA Practice Exam – Group 3 (101 – 150)

IP Connectivity, Services & Security

101.What command displays a router’s routing protocol process IDs and interfaces participating in OSPF?
A. show ip protocols
B. show ip ospf
C. show ip route ospf
D. show ip interface brief

102.In OSPF, what is the purpose of the “Router ID”?
A. It identifies each interface uniquely
B. It is the highest IP address on any interface unless manually set
C. It represents the AS number
D. It defines priority for DR election

103.Which OSPF state indicates that adjacency is complete?
A. Init
B. 2-Way
C. Full
D. ExStart

104.What happens if two routers have identical Router IDs in the same area?
A. OSPF fails to form adjacency
B. One router changes its ID automatically
C. Adjacency forms normally
D. Both become DR

105.Which command sets a router ID manually in process 1?
A. router ospf 1 → router-id 1.1.1.1
B. ip ospf router-id 1.1.1.1
C. ospf id 1.1.1.1
D. set router id 1.1.1.1

106.Which statement describes inter-VLAN routing on a multilayer switch?
A. Each VLAN requires a physical interface
B. It uses SVIs and the command ip routing
C. Trunk ports cannot be used
D. Routing must occur on a router only

107.What is the purpose of a default static route?
A. For routing protocols to advertise internal subnets
B. To route packets destined to unknown networks
C. To disable dynamic routing
D. To summarize multiple routes

108.Which command verifies a router’s default route?
A. show ip route 0.0.0.0
B. show ip route static
C. show ip route summary
D. show route default

109.A router learns identical prefixes from OSPF (AD 110) and static (AD 1). Which route is installed?
A. OSPF
B. Static
C. Both (load-balance)
D. None

110.Which term describes two equal-cost routes to the same destination?
A. Floating route
B. ECMP
C. Recursive route
D. Summary route

111.A router’s routing table contains both a connected and a static route to the same prefix. Which one wins?
A. Connected
B. Static
C. Longest prefix wins
D. Equal priority

112.What is the administrative distance of an OSPF-external route?
A. 90
B. 100
C. 110
D. 120

113.Which IPv6 address type is equivalent to IPv4 private addressing?
A. Global Unicast
B. Link-Local
C. Unique Local Address
D. Anycast

114.What is the IPv6 loopback address?
A. ::0
B. ::1
C. ::FFFF
D. FE80::1

115.Which IPv6 header field replaces the IPv4 TTL?
A. Flow Label
B. Hop Limit
C. Next Header
D. Traffic Class

116.What command displays IPv6 neighbor cache?
A. show ipv6 neighbors
B. show ipv6 nd
C. show ipv6 neighbor
D. show ipv6 arp

117.Which IPv6 process automatically generates a global address using MAC + prefix?
A. DHCPv6
B. SLAAC
C. Stateless NAT64
D. RA filtering

118.Which protocol is used by routers to advertise IPv6 prefixes for SLAAC?
A. DHCPv6-PD
B. Router Advertisement (ICMPv6)
C. RSVP
D. RIPng

119.A router interface has IPv6 address 2001:db8:10::1/64. What is the solicited-node multicast address?
A. FF02::1
B. FF02::2
C. FF02::1:FF00:1
D. FF02::1:FF10:1

120.Which FHRP allows multiple routers to share one virtual IP and provides load balancing?
A. HSRP
B. VRRP
C. GLBP
D. IRDP

121.Which feature verifies IP-to-MAC bindings to prevent spoofed ARP replies?
A. Port Security
B. Dynamic ARP Inspection
C. DHCP Snooping
D. BPDU Guard

122.Which command enables DHCP snooping for VLAN 50?
A. ip dhcp snooping vlan 50
B. dhcp snooping enable 50
C. ip dhcp trust 50
D. enable dhcp secure vlan 50

123.Which interface type should be configured as “trusted” for DHCP snooping?
A. Access port
B. Server uplink or trunk to DHCP server
C. End-user PC port
D. Voice port

124.What command shows DHCP snooping bindings?
A. show ip dhcp binding
B. show ip dhcp snooping binding
C. show dhcp binding secure
D. show ip helper

125.Which command limits learned MAC addresses on a port to 1 and shuts the port on violation?

```
interface Gi0/3 switchport port-security switchport port-security maximum 1
switchport port-security violation ?
```

A. protect
B. restrict
C. shutdown
D. disable

126.Which STP feature brings access ports to forwarding state immediately?
A. PortFast
B. Root Guard
C. BPDU Guard
D. Loop Guard

127.Which STP port state actively forwards frames?
A. Listening
B. Learning
C. Forwarding
D. Blocking

128.What is the function of the spanning-tree root bridge?
A. Acts as central reference for path cost calculation
B. Blocks redundant ports
C. Floods all VLAN frames
D. Disables BPDU exchange

129.Which STP feature disables a port receiving unexpected BPDUs?
A. BPDU Guard
B. Root Guard
C. Loop Guard
D. UDLD

130.Which command verifies the STP root for VLAN 10?
A. show spanning-tree vlan 10
B. show spanning-tree root
C. show stp vlan 10
D. show switch spanning

131.Which wireless topology connects clients through an AP?
A. Ad-hoc
B. Infrastructure
C. Mesh
D. WDS

132.Which authentication type uses 802.1X with a RADIUS server?
A. Open
B. WPA2-Personal
C. WPA2-Enterprise
D. PSK

133.Which security mode uses AES encryption and PSK authentication?
A. WPA Personal
B. WPA2 Personal
C. WEP
D. WPA3 Enterprise

134.A 2.4 GHz WLAN suffers from interference. Which adjustment reduces it?
A. Increase TX power
B. Move to 5 GHz
C. Enable WEP
D. Disable WMM

135.Which Cisco WLC mode allows an AP to keep serving clients if WAN link fails?
A. Local
B. FlexConnect
C. Monitor
D. Sniffer

136.Which command sets SSH version 2 on IOS device?
A. ip ssh version 2

B. enable ssh v2
C. crypto ssh 2
D. ssh-set 2

137.Which command creates a local user with privilege 15?
A. username admin secret Pass123
B. user add admin priv 15
C. create admin password Pass123
D. useradd admin secret Pass123

138.Which command secures VTY lines with local login?

```
line vty 0 4 ?
```

A. login local
B. transport input ssh
C. enable secret
D. username local

139.Which command encrypts plain-text passwords in running-config?
A. service password-encryption
B. enable password-encryption
C. encrypt passwords
D. security encryption enable

140.Which protocol provides confidentiality in IPsec VPNs?
A. AH
B. ESP
C. GRE
D. IKE

141.Which AAA component defines what resources a user may access?
A. Authentication
B. Authorization
C. Accounting
D. Auditing

142.Which protocol uses TCP port 49 for AAA communication?
A. RADIUS
B. TACACS+
C. LDAP
D. Kerberos

143.Which AAA protocol encrypts the entire payload of each packet?
A. RADIUS

B. TACACS+
C. Kerberos
D. LDAP

144.What is the difference between RADIUS and TACACS+ authentication?
A. RADIUS is Cisco proprietary; TACACS+ is open
B. TACACS+ encrypts full payload; RADIUS encrypts only password
C. Both use UDP 1812
D. RADIUS separates AAA functions

145.Which security technology limits MAC addresses per port?
A. DHCP Snooping
B. Port Security
C. DAI
D. STP

146.Which command displays currently learned secure MAC addresses?
A. show port-security address
B. show port-security interface
C. show mac address-table secure
D. show interface security

147.What is the default violation mode for Port Security?
A. protect
B. restrict
C. shutdown
D. drop

148.Which configuration enables syslog messages to a server at 192.168.1.50 with severity informational?

`logging host 192.168.1.50?`

A. logging trap informational
B. logging level informational
C. syslog informational on
D. logging server info

149.Which QoS mechanism marks packets with a DSCP value?
A. Classification
B. Marking
C. Shaping
D. Policing

150.Which command verifies an applied QoS policy on interface Gi0/0?
A. show policy-map interface Gi0/0
B. show qos policy Gi0/0
C. show service-policy Gi0/0
D. show ip qos interface

CCNA Practice Exam – Group 4 (151 – 200)

Routing, Switching, Security & Automation

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151. Refer to the exhibit.

Exhibit shows:

```
R1# show ip route C 10.10.10.0/24 is directly connected, Gig0/0 S* 0.0.0.0/0 [1/0] via 203.0.113.1
```

A host behind R1 cannot reach the Internet. What is the most likely cause?
A. Default route uses incorrect next-hop.
B. NAT is not configured on R1.
C. OSPF is disabled on Gig0/0.
D. The static route administrative distance is too low.

---

152. Refer to the exhibit.

Exhibit shows interface output:

```
R2# show interfaces Gig0/1GigabitEthernet0/1 is up, line protocol is down
```

What does this indicate?
A. Physical connection failure.
B. Data-link layer issue, likely encapsulation mismatch.
C. Interface is administratively down.
D. Cable unplugged.

---

153. Refer to the exhibit.

Exhibit shows:

```
SW1(config)# interface range gi0/1 - 2 SW1(config-if-range)# channel-group 1 mode activeSW1(config)# interface port-channel 1SW1(config-if)# switchport mode trunk
```

What protocol negotiates this EtherChannel?
A. PAgP

B. LACP
C. DTP
D. STP

---

154. Refer to the exhibit.

Exhibit shows VLAN output:

```
SW2# show vlan brief10 USERS active20 VOICE active99 NATIVE active
```

A new access port was configured for VLAN 20, but phones receive no IP address. What is missing?
A. switchport mode trunk
B. switchport voice vlan 20
C. ip helper-address
D. spanning-tree portfast

---

155. Refer to the exhibit.

Exhibit shows OSPF neighbor table:

```
R3# show ip ospf neighborNeighbor ID State Interface2.2.2.2 FULL/DR Gig0/04.4.4.4 2WAY/DROTHER Gig0/1
```

Why is neighbor 4.4.4.4 not full?
A. Hello/dead timers mismatch
B. Different area IDs
C. Non-DR/BDR on broadcast network only reach 2-Way state
D. MTU mismatch

---

156. Refer to the exhibit.

Exhibit shows:

```
R1(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2
```

What does this command accomplish?
A. Creates a static route for network 192.168.20.0 via next-hop 10.0.0.2.
B. Advertises route dynamically.
C. Creates a floating route.
D. Enables proxy ARP.

---

157. Refer to the exhibit.

Exhibit shows NAT config:

```
ip access-list standard INSIDE permit 10.10.0.0 0.0.255.255!interface Gig0/0 ip nat inside!interface Gig0/1 ip nat outside!ip nat inside source list INSIDE interface Gig0/1 overload
```

What NAT type is configured?
A. Static
B. Dynamic NAT with pool
C. PAT (NAT overload)
D. Twice NAT

---

158. Refer to the exhibit.

Exhibit shows:

```
Switch(config)# interface Gi0/1 Switch(config-if)# switchport port-security Switch(config-if)# switchport port-security maximum 1 Switch(config-if)# switchport port-security violation restrict
```

What occurs if a different MAC address appears on Gi0/1?
A. Port shuts down.
B. Frame dropped, log message generated.
C. Port err-disabled.
D. MAC added dynamically.

---

159. Refer to the exhibit.

Exhibit shows STP output:

```
SW1# show spanning-tree vlan 10Root ID Priority 32778 Address 0011.2233.4455Bridge ID Priority 32778 Address 00AA.BBCC.DDEE
```

Which device is the root bridge for VLAN 10?
A. SW1
B. SW2 (0011.2233.4455)
C. SW3 (00AA.BBCC.DDEE)
D. Both

---

160. Refer to the exhibit.

Exhibit shows:

```
SW1(config-if)# switchport trunk native vlan 999SW1(config-if)# switchport trunk allowed vlan 10,20
```

Why is VLAN 999 used here?
A. For voice traffic
B. For inter-VLAN routing
C. As security best practice to isolate untagged frames
D. As management VLAN

---

161. Refer to the exhibit.

Exhibit shows syslog messages:

```
%LINK-3-UPDOWN: Interface Gi0/1, changed state to up%PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred
```

What event triggered the second message?
A. New MAC learned on a secure port
B. Interface changed VLAN
C. Port shut down administratively
D. User cleared sticky MAC

---

162. Refer to the exhibit.

Exhibit shows CDP output:

```
Device ID: SW2Platform: WS-C2960XInterface: Gig0/1, Port ID: Gig0/2
```

Which protocol generated this output?
A. LLDP
B. DTP
C. CDP
D. LACP

---

163. Refer to the exhibit.

Exhibit shows fragment of OSPF configuration:

```
router ospf 10 network 10.10.10.0 0.0.0.255 area 0 network 172.16.0.0 0.0.255.255 area 1
```

What type of router is this?
A. Internal
B. Backbone
C. ABR
D. ASBR

---

164. Refer to the exhibit.

Exhibit shows JSON response from REST API:

```
{ "interface": { "name": "GigabitEthernet0/0", "description": "Uplink", "enabled": true }}
```

Which HTTP method likely retrieved this data?
A. POST
B. GET
C. PUT
D. DELETE

---

165. Refer to the exhibit.

Exhibit shows:

```
R1(config)# line vty 0 4 R1(config-line)# transport input ssh R1(config-line)# login local
```

What is the effect of this configuration?
A. Disables SSH access
B. Enables secure remote login using local credentials
C. Enables Telnet only
D. Uses AAA authentication

---

166. Refer to the exhibit.

Exhibit shows partial IPv6 config:

```
interface Gig0/0 ipv6 address 2001:db8:10::1/64 ipv6 enable
```

What additional command enables IPv6 routing?
A. ipv6 unicast-routing
B. ip routing v6 enable
C. enable ipv6
D. ipv6 enable-routing

---

167. Refer to the exhibit.

Exhibit shows IPv6 static route:

```
R2(config)# ipv6 route 2001:db8:2::/64 2001:db8:1::2
```

What is this route’s purpose?
A. Default route
B. Static route to remote network
C. Multicast route
D. Floating route only

---

168. Refer to the exhibit.

Exhibit shows HSRP output:

```
Vlan10 - Group 10 Local state is Standby Active router is 10.10.10.2, priority 110 Standby router is local, priority 100
```

Which router currently forwards client traffic?
A. Local router
B. Router 10.10.10.2
C. Both
D. Neither

---

169. Refer to the exhibit.

Exhibit shows ACL:

```
access-list 101 deny tcp any host 10.10.10.10 eq 23 access-list 101 permit ip any any
```

What is the effect?
A. Blocks Telnet to 10.10.10.10, permits all else
B. Blocks all TCP traffic
C. Blocks ICMP
D. Allows Telnet only

---

170. Refer to the exhibit.

Exhibit shows ping failure:

```
C:\>ping 10.10.10.1 Destination host unreachable
```

What layer is likely failing?
A. Layer 1
B. Layer 2
C. Layer 3
D. Layer 4

---

171. Refer to the exhibit.

Exhibit shows running-config:

```
username admin privilege 15 secret cisco123 service password-encryption
```

What is the result of this configuration?
A. All passwords displayed in clear text
B. Passwords are encrypted in configuration file
C. Passwords stored in type 7
D. Admin has privilege 1

---

172. Refer to the exhibit.

Exhibit shows Ansible playbook snippet:

```
- name: Configure VLAN 10 hosts: switches tasks: - ios_vlan: vlan_id: 10 name: USERS
```

Which automation tool syntax is this?
A. Puppet
B. Terraform
C. Ansible
D. Chef

---

173. Refer to the exhibit.

Exhibit shows:

```
Switch(config)# ip dhcp snoopingSwitch(config)# ip dhcp snooping vlan
10Switch(config)# interface gi0/24Switch(config-if)# ip dhcp snooping trust
```

What is the result?
A. DHCP packets blocked on VLAN 10
B. Switch acts as DHCP server
C. Only trusted uplink can send DHCP offers
D. All ports become trusted

---

174. Refer to the exhibit.

Exhibit shows interface counter output:

```
SW1# show interfaces Gig0/2 Input errors: 0 CRC: 100 Duplex: full
```

What is the likely cause?
A. Speed mismatch
B. Bad cable or interference
C. Duplex mismatch
D. VLAN mismatch

---

175. Refer to the exhibit.

Exhibit shows part of syslog configuration:

```
logging host 192.168.1.10 logging trap warnings
```

Which messages are sent?
A. Severity 0–4
B. Severity 0–5
C. Severity 0–7
D. Severity 4–7

---

176. Refer to the exhibit.

Exhibit shows:

```
Router(config)# ip dhcp pool USERS network 10.10.10.0 255.255.255.0 default-router 10.10.10.1 dns-server 8.8.8.8
```

What is required for clients in a different VLAN to obtain addresses from this pool?
A. ip helper-address on SVI
B. NAT configuration
C. DHCP snooping trust
D. Static binding

---

177. Refer to the exhibit.

Exhibit shows:

```
R1(config)# router ospf 1 R1(config-router)# passive-interface Gig0/0
```

What is the effect?
A. OSPF stops advertising network 10.0.0.0
B. OSPF sends Hellos but no LSAs
C. No OSPF Hellos sent out Gig0/0, but network still advertised
D. Interface removed from OSPF area

---

178. Refer to the exhibit.

Exhibit shows REST API request:

```
POST /restconf/data/interfaces/interface=GigabitEthernet0%2F0
```

What operation is being performed?
A. Read existing data
B. Delete resource
C. Create new configuration
D. Retrieve statistics

---

179. Refer to the exhibit.

Exhibit shows JSON fragment:

```
{ "device": { "hostname": "R1", "interfaces": [ {"name": "Gig0/0"}, {"name": "Gig0/1"} ] }}
```

What data type is represented by the square brackets `[ ]`?
A. Dictionary
B. Array (list)
C. Object
D. Map

---

180. Refer to the exhibit.

Exhibit shows two routes:

```
R1# show ip route 10.10.20.0S 10.10.20.0/24 [1/0] via 192.168.1.2 O 10.10.20.0/24 [110/2] via 192.168.1.3
```

Which route is installed in the routing table?
A. Static
B. OSPF
C. Both (ECMP)
D. Neither

---

181. Refer to the exhibit.

Exhibit shows interface summary:

```
Gig0/1 connected trunk a-full a-1000 10,20
```

What does the “a-” prefix indicate?
A. Administratively assigned
B. Auto-negotiated
C. Active mode
D. Alternate

---

182. Refer to the exhibit.

Exhibit shows switch stack output:

```
Switch# show switchSwitch 1 Role: ActiveSwitch 2 Role: Standby
```

What does this output represent?
A. EtherChannel
B. StackWise redundancy
C. VTP domain
D. STP root bridge

---

183. Refer to the exhibit.

Exhibit shows ACL applied to interface:

```
interface Gig0/1 ip access-group 101 in
```

Packets matching a deny statement are dropped. How can these drops be monitored?
A. Use the log keyword in ACL entry
B. Apply ACL outbound
C. Enable debugging IP packet
D. Configure syslog level 7

---

184. Refer to the exhibit.

Exhibit shows:

```
Switch(config-if)# spanning-tree portfast Switch(config-if)# spanning-tree bpduguard enable
```

What is the combined effect?
A. Port forwards instantly but shuts down if BPDU received
B. Disables STP
C. Blocks all BPDUs
D. Forwards BPDUs to root

---

185. Refer to the exhibit.

Exhibit shows wireless config:

```
SSID: Guest-WiFi Security: WPA2-PSK Passphrase: guest123
```

What encryption method is used?
A. TKIP
B. AES-CCMP
C. RC4
D. MD5

---

186. Refer to the exhibit.

Exhibit shows:

```
interface Vlan10 ip address 10.10.10.1 255.255.255.0 ip helper-address 10.0.0.5
```

What is the purpose of this command?
A. Forwards DHCP broadcasts to server 10.0.0.5
B. Assigns static IP to DHCP server
C. Enables DNS resolution
D. Provides default route

---

187. Refer to the exhibit.

Exhibit shows:

```
Router(config)# snmp-server community public RO
```

What is the security risk?
A. Allows remote config changes
B. Uses clear-text community string
C. SNMP disabled
D. Encrypts traps only

---

188. Refer to the exhibit.

Exhibit shows an NTP status:

```
R1# show ntp statusClock is synchronized, stratum 3
```

What does “stratum 3” mean?
A. R1 is 3 hops away from a reference clock

B. NTP unsynchronized
C. R1 is master
D. Drift detected

---

189. Refer to the exhibit.

Exhibit shows:

```
Switch(config)# banner motd # Unauthorized access prohibited #
```

When does this banner appear?
A. Before login prompt
B. After login
C. When SSH starts
D. During enable mode

---

190. Refer to the exhibit.

Exhibit shows:

```
Switch# show etherchannel summaryGroup 1 Gi0/1(GP) Gi0/2(P)
```

What does “P” indicate?
A. Port in use in the bundle
B. Passive mode
C. Port suspended
D. Port err-disabled

---

191. Refer to the exhibit.

Exhibit shows:

```
Switch(config)# interface Gi0/3Switch(config-if)# spanning-tree guard root
```

What is the effect of this configuration?
A. Prevents port from becoming root port
B. Makes port root
C. Disables BPDUs
D. Forces VLAN 1 as root

---

192. Refer to the exhibit.

Exhibit shows HTTP request header:

```
POST /api/v1/interfaces HTTP/1.1Host: 10.0.0.1Content-Type:
application/jsonAuthorization: Bearer token123
```

Which method is used for API authentication?
A. Token-based (Bearer)
B. Basic auth
C. API key
D. Cookie session

---

193. Refer to the exhibit.

Exhibit shows:

```
Switch# show ip interface briefInterface IP-Address OK? Method Status ProtocolVlan1
192.168.1.2 YES manual up up
```

What does “up/up” indicate?
A. Interface operational
B. Physical layer failure
C. Disabled VLAN
D. Layer 3 mismatch

---

194. Refer to the exhibit.

Exhibit shows routing table:

```
Gateway of last resort is not set
```

What is the result?
A. No default route configured
B. NAT disabled
C. OSPF not running
D. Routing loop detected

---

195. Refer to the exhibit.

Exhibit shows:

```
Switch(config)# ip arp inspection vlan 10
```

What must be configured for this feature to function?
A. DHCP snooping
B. PortFast

C. BPDU Guard
D. NAT

---

196. Refer to the exhibit.

Exhibit shows Python code:

```
import requestsresponse = requests.get("http://192.168.1.1/api/v1/interfaces")print(response.status_code)
```

What does `status_code 200` indicate?
A. Request failed
B. Successful request
C. Unauthorized
D. Redirect

---

197. Refer to the exhibit.

Exhibit shows:

```
Switch(config)# interface range Gi0/1 - 2Switch(config-if-range)# shutdown
```

What is the result?
A. Both interfaces administratively disabled
B. One interface disabled
C. VLANs deleted
D. Interface speed reduced

---

198. Refer to the exhibit.

Exhibit shows interface config:

```
interface Gig0/1 description LINK_TO_R2 ip address 10.0.0.1 255.255.255.252
```

What type of link is this?
A. Loopback
B. Point-to-point
C. Transit
D. Broadcast

---

199. Refer to the exhibit.

Exhibit shows JSON snippet:

```
{ "hostname": "Switch1", "vlans": {"10": "Users", "20": "Voice"}}
```

What structure is `"vlans"` here?
A. Object with key-value pairs
B. Array
C. List
D. Integer

---

200. Refer to the exhibit.

Exhibit shows CLI:

```
Switch# delete vlan.dat
```

What is the effect of this command?
A. Removes VLAN database and resets VLAN configuration
B. Deletes running-config
C. Erases flash memory
D. Resets STP